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1. System Overview - Kya Ready Hai

CSC Smart Seva ek WhatsApp-based digital seva automation system hai jo aapke customers ko 14 sarkari services WhatsApp chat ke through deta hai. System abhi tak fully built aur tested hai: 25 n8n workflows, 22 database tables, ek admin portal, aur complete backup files. Ye guide step-by-step batati hai ki ise production mein live kaise karein - server se lekar WhatsApp production aur live payments tak.

Abhi system sandbox mein MOCK mode par chal raha hai. Iska matlab hai ki WhatsApp messages aur payments dono simulate ho rahe hain, lekin poora logic - intent detection, field validation, document collection, payment verification, receipts aur operator tasks - real hai. Live karne ke liye sirf 4 cheezein badalni hain: ek server, ek domain, Meta WhatsApp production credentials aur Razorpay live keys.

1.1 Current Assets

Asset	Detail	Status
n8n Workflows	25 workflows (CSC 01-25), hub-and-spoke architecture	Built + tested
Data Tables	22 tables, 237+ seed rows (fields, docs, pricing, config)	Populated
Admin Portal	Next.js dashboard - applications, tasks, payments, settings	Built + verified
Supabase Backup	workflows-backup.json, datatables-backup.json, credentials	Ready
Migration Scripts	import-datatables.js + activation scripts	Ready
Operator Guide	CSC_Services_Operator_Guide.pdf (14 services)	Delivered

Table 1: Ready assets inventory

Sabse important baat: services ke beech koi hardcoding nahi hai. Har service ka fields, documents aur pricing database tables (service_engine config) se aata hai. Isliye aap portal se pricing change kar sakte ho, naya field add kar sakte ho, aur workflow ko touch karne ki zaroorat nahi padti. Yehi design production mein aapko sabse zyada time bachayega.

1.2 Live Architecture (Target)

Live hone par system aise chalega: Customer ka WhatsApp message Meta Cloud API se aapke server par bane n8n webhook par aayega. n8n intent samjhega, fields collect karega, documents lega, aur payment link banayega. Payment confirm hone par application finalize hoti hai, receipt jaata hai, operator task banta hai aur admin portal mein dikhne lagta hai. Operator portal se approve karta hai aur certificate deliver hota hai. Saara data Supabase Postgres mein rehta hai jo n8n aur portal dono share karte hain.

Component	Role	Hosting Target
Meta WhatsApp Cloud API	Message receive/send (official channel)	Meta cloud (free)

Component	Role	Hosting Target
n8n (25 workflows)	Bot brain - conversation, validation, payments	India VPS (Docker/bun)
Supabase Postgres	Single source of truth - 22 tables	Supabase cloud
Admin Portal (Next.js)	Applications, tasks, services, settings manage	Same VPS (PM2) / Vercel
Razorpay	UPI/cards collection + webhook verification	Razorpay cloud

Table 2: Production architecture components

2. Go-Live Roadmap - 8 Steps

Poora launch 3-5 din ka kaam hai, agar Meta business verification fast ho jaye. Neeche wala roadmap follow karein - sequence important hai kyunki har step previous step par depend karta hai. Sabse pehle server aur database ready karein, phir domain lagayen, uske baad WhatsApp webhook register hoga (HTTPS zaroori hai), aur last mein live payments on honge.

Step	Kaam	Time	Dependency
1	VPS server kharidna aur n8n install karna	2-3 ghante	-
2	Supabase Postgres setup + data restore	1-2 ghante	Step 1
3	Domain + HTTPS (nginx + SSL)	1 ghanta	Step 1
4	Workflows import + activate + DB switch	2 ghante	Step 2, 3
5	Meta WhatsApp production credentials	1-3 din (verification)	Step 3
6	Razorpay live keys + webhook	2-4 ghante	Step 3
7	Admin portal deploy + password change	1 ghanta	Step 2
8	End-to-end testing + soft launch	1 din	Sab steps

Table 3: Go-live sequence with time estimates

Golden Rule

Kabhi bhi seedha live par switch na karein. Pehle TEST mode mein 10-15 real applications chalayen (apne number se), phir PAYMENT_MODE aur WhatsApp credentials live karein. Soft launch strategy: pehle 20 trusted customers, phir full public.

2.1 Monthly Cost Estimate

Item	Option A (Budget)	Option B (Recommended)
VPS 2GB RAM	Hostinger/DO Bangalore: Rs 400-700/mo	DO Bangalore 2GB: Rs 800/mo
Supabase	Free tier (500MB, enough for start)	Pro Rs 1,500/mo (jab data badhe)
Domain .in/.com	Rs 100-800/year	Rs 800/year
WhatsApp Cloud API	Free (1000 service convos/mo)	Free + usage-based after limit
Razorpay	2% per transaction (UPI 0%)	Same
n8n license	Self-host free (Community)	Self-host free
Approx total	Rs 500-700/month	Rs 900-1,200/month

Table 4: Estimated monthly running cost (INR)

Shuruat mein Option A kaafi hai. 500 applications per month tak free Supabase tier comfortably chalega. WhatsApp Cloud API ka pricing conversation-based hai - customer-initiated service conversations (24-hour window) ke first 1,000 per month free hain, jo ek naye CSC ke liye kaafi se kaafi zyada hai. Baad mein jab volume badhega, Meta ka per-conversation charge lagbhag Rs 0.30-0.80 aata hai.

3. Server Setup - VPS India

Server ke liye India region choose karna zaroori hai kyunki WhatsApp aur government portals dono ka latency India se best rehta hai, aur Razorpay webhooks bhi fastest yahin aate hain. DigitalOcean Bangalore, AWS Mumbai, ya Hostinger India - teeno theek hain. Minimum 2GB RAM chahiye kyunki n8n ko workflows chalane ke liye memory chahiye aur Next.js portal bhi wahi chalega. Ubuntu 22.04 LTS LTS image select karein aur SSH key setup karke login karein.

3.1 Base Installation

SSH login ke baad ye commands ek-ek karke chalayen. Ye Node.js 20, bun runtime, n8n, nginx aur PM2 install karega - poora stack ek hi server par:

```
apt update && apt upgrade -y

curl -fsSL https://deb.nodesource.com/setup_20.x | bash - && apt install -y nodejs
nginx

npm install -g bun pm2

mkdir -p /opt/csc && cd /opt/csc && bun install n8n@latest
```

n8n ko bun se install karna intentionally kiya gaya hai - pichle testing mein npm se install par n8n OOM (out of memory) hota tha jab 25 workflows ek saath active the, jabki bun runtime memory efficiently handle karta hai. n8n version 2.40.7 verified hai hamare workflows ke saath, to bilkul naya version aane par bhi pehle staging mein test karein.

3.2 systemd Service (Auto-Start)

Server reboot hone par n8n khud start hona chahiye. Iske liye systemd service banayen:

```
nano /etc/systemd/system/n8n.service
```

```
[Unit] Description=n8n CSC | After=network.target
[Service] Type=simple | User=root | WorkingDirectory=/opt/csc
Environment=N8N_HOST=csc.yourdomain.in
Environment=N8N_PORT=5678 | Environment=N8N_PROTOCOL=https
Environment=WEBHOOK_URL=https://csc.yourdomain.in/
Environment=GENERIC_TIMEZONE=Asia/Kolkata | Environment=TZ=Asia/Kolkata
Environment=DB_TYPE=postgresdb
Environment=DB_POSTGRESDB_HOST=aws-0-ap-south-1.pooler.supabase.com
Environment=DB_POSTGRESDB_DATABASE=postgres | Environment=DB_POSTGRESDB_PORT=5432
Environment=DB_POSTGRESDB_USER=postgres.xxx | Environment=DB_POSTGRESDB_PASSWORD=xxxx
ExecStart=/usr/local/bin/bun x n8n start
Restart=always | RestartSec=5
[Install] WantedBy=multi-user.target
```

```
systemctl daemon-reload && systemctl enable n8n && systemctl start n8n
```

Timezone Important

GENERIC_TIMEZONE=Asia/Kolkata aur TZ dono set karein. Reminder system (CSC 21) ke quiet hours (21:00 se 09:00) aur daily admin digest (CSC 23) ke schedules isi timezone se calculate hote hain. Galat timezone par reminders raat ko 2 baje jayenge.

4. Database - Supabase Postgres Migration

Abhi sandbox mein n8n apna data SQLite file mein rakhta hai. Production mein SQLite theek nahi hai - backup mushkil, concurrent writes slow, aur portal share nahi kar sakta. Supabase Postgres pe se hi hamara target tha isliye migration files already ready hain. Supabase account banakar naya project create karein (region: ap-south-1 Mumbai), aur Project Settings se connection string copy karein.

4.1 Restore Steps

Migration folder (download/supabase-migration/) ke saare files server par /opt/csc/migration/ mein copy karein. Phir ye order follow karein:

- **Step A:** n8n ko pehli baar DB_TYPE=postgresdb ke saath start karein - ye automatically saare n8n internal tables bana dega.
- **Step B:** node import-datatables.js chalayen - ye 22 CSC tables (applications, service_fields, service_documents, service_pricing, system_config etc.) Supabase public schema mein banayega aur 237+ seed rows daalega.
- **Step C:** n8n UI (Settings > Data Tables) mein confirm karein ki tables dikh rahe hain.

- **Step D:** workflows-backup.json import karein - n8n UI se (Import from File) ya CLI se: `n8n import:workflow --input=workflows-backup.json`
- **Step E:** Credentials manually re-enter karein (WhatsApp token, Razorpay keys) - security ke liye credentials backup encrypted hai.
- **Step F:** Sab 25 workflows activate karein. Import ke baad workflows deactivated hote hain - ye n8n ka known behaviour hai. Ek-ek karke UI se toggle karein ya official API use karein.

Activation Pitfall

SQLite me directly `active=1` set karne se kaam NAHI chalta - n8n cache follow karta hai. Official REST API use karein: `POST /api/v1/workflows/{id}/activate` with header `X-N8N-API-KEY`. Isi se activation guaranteed hoti hai. Activate-all script `scripts/phase-build/activate-all.sh` me ready hai.

4.2 Verify Migration

Migration ke baad ye 3 checks karein. Pehla: n8n UI mein koi bhi workflow open karke Data Table node dekhein - usme `service_fields` table dikhna chahiye. Doosra: Supabase SQL Editor mein `SELECT COUNT(*) FROM service_catalog` chalayen - 16 rows aani chahiye. Teesra: `system_config` table mein `PAYMENT_MODE=MOCK` abhi bhi hona chahiye (live switch baad mein hoga).

Check	Expected Result
<code>SELECT COUNT(*) FROM service_catalog</code>	16
<code>SELECT COUNT(*) FROM service_fields</code>	99
<code>SELECT COUNT(*) FROM service_documents</code>	54
<code>SELECT config_value FROM system_config WHERE config_key = 'APP_NUMBER_PREFIX'</code>	CSC-2026-
n8n workflows active	25 of 25

Table 5: Post-migration verification queries

5. Domain + HTTPS Setup

Meta WhatsApp Cloud API webhook ke liye HTTPS URL mandatory hai - HTTP par webhook register nahi hota. Isliye ek domain kharidna (GoDaddy/Namecheap/Hostinger se .in ya .com) aur usse VPS ke IP par point karna pehla infrastructure kaam hai. DNS A record mein @ aur dono subdomains add karein: `bot.yourdomain.in` (n8n webhook) aur `admin.yourdomain.in` (portal).

DNS propagate hone ke baad (15-30 minute) nginx reverse proxy setup karein. nginx port 80/443 par sunta hai aur traffic internally n8n (port 5678) aur portal (port 3000) mein baant deta hai:

```
server { listen 80; server_name bot.yourdomain.in;
location / { proxy_pass http://localhost:5678;
proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr;
proxy_set_header X-Forwarded-Proto https; } }
```

```
apt install certbot python3-certbot-nginx
certbot --nginx -d bot.yourdomain.in -d admin.yourdomain.in
```

Certbot automatically Let's Encrypt SSL certificate laga dega aur HTTP ko HTTPS par redirect kar dega. Certificate 90 din ka hota hai aur certbot renewal cron khud set karta hai. Ek baar <https://bot.yourdomain.in/healthz> open karke confirm karein ki n8n ka response aa raha hai.

6. WhatsApp Production (Meta WABA)

Ye sabse important aur sabse time-taking step hai kyunki Meta ki business verification mein 1-3 din lag sakte hain. SandBox se production jaane ke liye ye cheezein chahiye: Meta Business Portfolio, ek verified business, WhatsApp Business Account (WABA), aur ek phone number jo WhatsApp par registered NAHI ho (na app mein, na kisi aur account mein). Number naya kharidna best hai - purana number agar personal WhatsApp par hai to pehle usse delete karna hoga.

6.1 Meta Setup Checklist

Step	Action	Note
1	developers.facebook.com par app banayein (type: Business)	Free
2	WhatsApp product add karein + WABA create karein	Auto
3	Business verification documents submit karein	1-3 din
4	Phone number register karein (display name approve karayein)	Rental/new SIM
5	System User banakar permanent token generate karein	Never expiring
6	phone_number_id + waba_id copy karein	App dashboard
7	Webhook subscribe karein + verify token set karein	messages field ON

Table 6: Meta WhatsApp production setup sequence

6.2 Credentials Portal Mein Dalna

Token milne ke baad Admin Portal > Settings > WhatsApp Cloud API section mein ye 4 values bharein (ya seedhe Supabase system_config table mein update karein). n8n har message par fresh config padhta hai, isliye restart ki zaroorat nahi:

Config Key	Kya Bharna Hai	Example
WHATSAPP_PHONE_NUMBER_ID	Meta dashboard se phone number ID	123456789012345
WHATSAPP_ACCESS_TOKEN	System User permanent token	EAAG...long-string
WHATSAPP_API_VERSION	Graph API version	v21.0
ADMIN_WHATSAPP_NUMBER	Aapka operator WhatsApp (91 prefix)	9198XXXXXX99

Table 7: WhatsApp system_config values

Webhook URL Meta app dashboard mein yeh dalein: <https://bot.yourdomain.in/webhook/csc-router> Verify token field mein koi bhi string dein (system mein csc_verify_2026 set hai) aur messages webhook field subscribe karein. Jab bhi customer message bhejega, Meta usse is URL par POST karega aur CSC 01 router workflow usse process karega.

6.3 24-Hour Window aur Templates

WhatsApp ka sabse bada rule: aap customer ko sirf 24-hour window ke andar free-form message bhej sakte ho - window tab khulti hai jab customer ne aapko message kiya ho. Hamara poora bot flow customer-initiated hai isliye yahan koi problem nahi aayegi. Lekin reminders (CSC 21) customer-initiated window ke bahar ja sakte hain - payment reminder bhejne ke liye approved message template use karna padega. Meta dashboard se 3 templates banayein:

- **payment_reminder:** "Namaste {{1}}, aapki {{2}} application pending hai. Payment complete karne ke liye: {{3}}"
- **status_update:** "Aapki application {{1}} ka status: {{2}}. Detail ke liye reply karein."
- **certificate_ready:** "Badhai ho {{1}}! Aapka {{2}} ready hai. Download: {{3}}"

Quality Rating

Naye numbers par Meta "Limited Access" (250 unique customers/day) deta hai. 30 din tak achha quality rating maintain karne par limit badhti hai. Spam na karein - REMINDER_MAX_PER_APP=3 config already har application par maximum 3 reminders limit karta hai.

7. Razorpay Live Payments

Payments live karne ke liye Razorpay dashboard se activation complete karein: business KYC (PAN, GST ya Shop Act, bank account) submit karein. Approval ke baad dashboard ke Live Mode tab se Key ID aur Key Secret milenge. Phir Admin Portal Settings mein ye karein:

Config Key	Value	Kahan Se
PAYMENT_MODE	RAZORPAY_LIVE	Manual (MOCK se change karein)

Config Key	Value	Kahan Se
RAZORPAY_KEY_ID	rzp_live_XXXXXXX	Dashboard - API Keys
RAZORPAY_KEY_SECRET	(secret string)	Dashboard - API Keys (ek baar dikhta hai)
RAZORPAY_WEBHOOK_SECRET	csc_webhook_secret_2026	Same rakhna hai - workflows isi se verify karte hain

Table 8: Payment config keys

Razorpay dashboard mein webhook bhi register karein: URL `https://bot.yourdomain.in/webhook/csc-payment-callback`, event `payment.captured`, aur secret `csc_webhook_secret_2026`. CSC 12 workflow har webhook par HMAC-SHA256 signature verify karta hai - galat secret wale fake requests reject ho jayengi. Ye security layer already built aur tested hai (demo mein CSC-2026-EAUM1N application isi flow se PAID hui thi).

Go-Live Test Pattern

Live keys dalne ke baad pehla test apne hi number se karein: poora PAN flow complete karein aur Rs 1 test product par real UPI payment karein. Confirm karein ki (1) Razorpay dashboard mein payment dikhe, (2) application status PAID ho, (3) receipt WhatsApp par aaye, (4) operator task bane. Sab 4 pass hone par hi public launch karein.

8. Admin Portal Deployment

Admin portal Next.js application hai jo aapko sab kuch ek jagah manage karne deti hai: applications (form data, documents, status timeline), operator tasks (approve/reject), 16 services ki config (fields, documents, pricing), payments ledger, live conversations aur system settings. Portal Supabase Postgres se directly connect hota hai - wahi database jo n8n use karta hai. Isliye dono ka data hamesha synced rehta hai.

Portal deploy karne ke 2 options hain. Option A (recommended): wahi VPS jahan n8n chal raha hai - extra cost nahi. Option B: Vercel pe free hosting - easy but DATABASE_URL public internet se Supabase ko hit karti hai (SSL enforced hai to safe hai). Option A ke steps:

- Portal folder `/opt/csc/portal/` mein copy karein (ya git clone karein).
- `.env` file banayein: DATABASE_URL (Supabase pooler URL), ADMIN_PASSWORD (strong password), PORTAL_SECRET (random 32 chars).
- `bun install && npx prisma generate && bun run build` chalayen.
- `pm2 start npm --name csc-portal -- run start (port 3000)` chalayen.
- nginx mein `admin.yourdomain.in` server block add karein (`proxy_pass http://localhost:3000`).

First Login

Portal ka default password csc-admin-2026 hai (sirf pehli login ke liye). Login ke turant baad .env me ADMIN_PASSWORD change karke pm2 restart csc-portal chalayen. Production me default password kabhi na chhodein.

Portal ke andar saara data n8n ke saath same tables se aata hai. Operator jab task approve karta hai to portal application_status_history aur audit_log mein entry likhta hai - yaani poora audit trail preserved rehta hai. Application status flow portal mein is tarah manage hota hai: DRAFT > SUBMITTED > PAID > IN_PROGRESS > APPROVED > DELIVERED (REJECTED bhi kahin se bhi possible hai).

9. Pre-Launch Checklist

Public launch se pehle neeche wale 12 tests sequentially pass karne zaroori hain. Har test ek real scenario simulate karta hai. Inme se koi bhi fail ho to launch postpone karein aur fix karein - customer ke saamne broken bot brand image kharab karta hai.

#	Test	Expected
1	Naye number se "hi" message	Welcome + services list aaye
2	"PAN card banwana hai" bolo	Service confirm + price dikhaye (Rs 166)
3	Galat phone number type karo (5 digit)	Validation error message aaye
4	Saare 7 fields + 3 documents bhejo	Payment link aaye
5	Payment link se UPI payment karo	Receipt + confirmation aaye
6	Razorpay dashboard check karo	Payment captured dikhe
7	Portal > Applications kholo	Nayi application PAID status par dikhe
8	Portal > Tasks me approve karo	Status APPROVED ho + history bane
9	"status" message customer se	Bot approved status bataye
10	Payment ke bina 30 min wait karo	Reminder aaye (max 3, quiet hours respect)
11	Admin number par daily digest	Roz summary aaye
12	Unknown/galat message ("asdfgh")	Bot gracefully help bataye, crash na ho

Table 9: 12-point pre-launch test suite

Test 10 ke liye note karein: reminder turant nahi aata - CSC 21 har ghante check karta hai ki kaunsi applications 24 ghante se pending hain, aur unme se qualified par reminder bhejta hai. Testing ke waqt REMINDER_MAX_PER_APP ko temporarily 1 rakh sakte ho taaki spam na ho.

10. Maintenance & Monitoring

Live hone ke baad daily 10 minute ka routine kaafi hai. Subah portal kholke dekhein: kal kitni applications aayi, kitni PAID hui, pending tasks kitne hain. Conversations page par wo customers dekhein jinke state PAYMENT ya DOCS par atke hain - unhe manually reminder bhejna (ya portal se status change karna) chahiye. Weekly ek baar Payments page se Razorpay dashboard reconcile karein - total collected match hona chahiye.

10.1 Daily & Weekly Routine

Frequency	Task	Kahan
Daily subah	Pending tasks approve/reject karo	Portal - Operator Tasks
Daily subah	Atki conversations dekho	Portal - Conversations
Daily	n8n executions me errors check karo	n8n UI - Executions (Status=Error)
Weekly	Payments reconcile (Razorpay vs Portal)	Portal - Payments
Weekly	Server disk + memory check karo	SSH: df -h, free -m
Monthly	Supabase backup download karo	Supabase - Backups
Monthly	WhatsApp quality rating dekho	Meta Business Manager

Table 10: Maintenance routine

10.2 Backup Strategy

Teen layers ki backup rakhein. Pehla: Supabase ka built-in daily backup (Pro plan mein automatic, free tier mein manual download). Doosra: har hafte n8n ka export - workflows ko n8n CLI se export karke ya UI se download karke ek folder mein rakhein (aaj ke backup files already isi format mein hain). Teesra: system_config aur service config ka snapshot - portal ke Settings page se JSON copy karke store karein. Server crash ho jaaye to 2-3 ghante mein poori system naye VPS par restore ho jayega.

Ek aur cheez: workflows ko edit karne se pehle hamesha pehle export karke backup lein. 25 workflows ka hub-and-spoke design hai - koi bhi workflow change karte waqt uske callers ka dhyan rakhein (CSC 20 sender sab use karte hain, CSC 01 router sabse pehle hai). Ek waqt mein sirf ek workflow change karein aur turant test karein. Merge na karein - alag workflows isliye hain taaki ek service ka fix doosri service ko tod na sake.

10.3 Common Issues & Fixes

Problem	Likely Cause	Fix
Customer ko reply nahi aa raha	Token expire / webhook unsubscribed	Meta dashboard check, naya token Settings me dalo
Payment aaya but status PAID nahi	Webhook secret mismatch	RAZORPAY_WEBHOOK_SECRET dono jagah same rakho
Reminders nahi ja rahe	Template approved nahi / quiet hours	Meta templates check, REMINDER_QUIET config dekho
Portal login nahi ho raha	ADMIN_PASSWORD changed/env missing	.env check karo, pm2 restart csc-portal
n8n OOM / slow	2GB kam pad raha	Swap add karo ya 4GB VPS upgrade
Executions me red errors	Data table name changed	Error log kholo, node input dekho

Table 11: Troubleshooting quick reference

Is guide ke saath aapke paas poora launch kit hai: server setup commands, database restore steps, WhatsApp aur payment production credentials ka setup, portal deployment, testing checklist aur maintenance routine. Launch ke baad pehle mahine specifically dhyan rakhein ki customers ka experience smooth ho - bot ki language simple Hinglish mein hai, validation errors helpful hain, aur reminder limits anti-harassment ke saath tuned hain. Jab volume badhe, naye services add karna sirf database rows ka kaam hai - koi naya workflow nahi banana padega.